

Assignment Asset Pricing*

FEM21003, Econometrics & Management Science

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1 Introduction

Machine learning techniques have been proven to be quite useful in empirical asset pricing over the last decade. Empirical research has uncovered many firm and stock characteristics that are relevant for return prediction, portfolio formation and factor creation. So many, that this situation is commonly referred to as the “factor zoo”, a term originally coined by Cochrane (2011). Surveying the literature, Jensen et al. (2023) present an overview of 153 factors and show that many factors still yield a significant risk premium.¹ Many of these factors are quite related, so dimension reduction seems a relevant method.

A popular technique to reduce the dimension of a large set of random variables is Principal Component Analysis (PCA). Applied to returns on individual stocks or portfolios, PCA can be seen as minimizing the time-series variation that is left unexplained by the principal components. Kelly et al. (2019) propose an extension where the exposures to the principal components can vary over time. The exposures become a function of stock or portfolio characteristics. They call their method Instrumented PCA (IPCA) because the role of characteristics can be interpreted as instruments.

In this assignment you will investigate how IPCA performs relative to standard PCA and *a priori* constructed factor models in the style of Fama and French (1993, 2015) based on

*For questions, please use the forum available on Canvas.

¹These factors are available at <https://jkpfactors.com/>.

a portfolio sort of your own choice. The assignment consists of four questions that help you structure your investigation and a fifth question to conclude.

You should write a short report of 1 page per question. The answer to each question should consist of a graph or table and a set of bullet points that summarize your choices with regard to data and hyper parameters, and your main insights. You should finish with a sixth page with the overall conclusion, suggestions for further research, reflection on the peer review, references and a declaration of the software and packages you used, including your use of generative AI. You do not need to repeat the methods you use, as they are either covered by the lecture slides or the articles.

2 Design and planning

Date	Activity
Tuesday, 8/9	Introduction Assignment
Thursday, 1/10, 23h59	Registration in Canvas, submission report with questions 1–3
Friday, 2/10, 12h59	Reports available for peer review
Wednesday, 7/10, 12h59	Submission peer review
Wednesday, 7/10, 14h59	Release peer review
Thursday, 15/10, 23h59	Submission final version

3 Organisation

The assignment is organized as follows

- You have to work in teams of four. Team members should follow the same course (i.e. all in FEM11008 or all in FEM21003).
- You have to register yourself as a group on Canvas in the group set “AP Assignment”. You can pick any free group number.
- Submission on Canvas should be done by the student with the lowest student number.

- Giving proper feedback is mandatory. *We will deduct points if you do not give proper feedback.*
- The feedback is not reciprocal, so you will receive feedback from a group other than the one that you give feedback to.
- The sample period that you should use for your data is prescribed. On Canvas, an Excel file with student numbers, start dates and end dates is available. You should use the start and end date from the student with the lowest student number.

4 Data and preparation

- Select a set of 5×5 double-sorted portfolios and download monthly returns from Kenneth French's data library.² The first sort is always size, the second sort can be based on Book-to-Market, Operating Profitability, Investment, Momentum, Short-Term Reversal, or Long-Term Reversal.
- The files with the double sorted portfolios contain a set of portfolio characteristics, following after the series of annual equal-weighted returns. The number of firms per portfolio is not a useful instrument for IPCA, but all others are. The characteristics are reported such that they are available at the beginning of the month (denoted by $\mathbf{z}_t$ in the paper), whereas the reported returns correspond with the end of the month (denoted by $\mathbf{r}_{t+1}$ in the paper).
- The reported characteristics can be seen as 'raw'. You can directly use them in the analysis, but perhaps some transformations make them more useful. Kelly et al. (2019) use a rank transformation: for each point in time, they calculate the rank of the observation of a characteristic (so in the cross-section), and transform this to the $[-0.5, 0.5]$ domain. When working with individual assets, that also limits the influence of outliers, but this is less of a concern when working with portfolios. You can make your own choices here, and you can also include several transformations in the analysis (as long as there is no perfect correlation between transformations).

²Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

- You can use the portfolio returns to create characteristics. For example, you can construct a momentum-like characteristic by summing past returns. The characteristic for portfolio i at time t can then be the sum of the past returns from $t - 12$ up to $t - 2$, $z_{i,t} = \sum_{\tau=2}^{12} r_{i,t-\tau}$. Alternatively, you can use these returns to calculate a volatility measure.
- Make sure you have at least 5 characteristics. Therefore, if you select the portfolios where the second sort is on Momentum, Short-Term Reversal, or Long-Term Reversal, you have to create several (transformations of) characteristics of your own.
- Download the three Fama-French factors that correspond to the portfolio sorts you picked. The first factor is always the excess market return (labelled "Mkt-Rf"), the second factor is always SMB, and the third factor depends on the sort you picked. The factors HML, RMW and CMA are included in the "Fama-French 5 Factor" file. This file also contains the series for the risk-free rate (labelled "RF"). There are separate files for the factors related to momentum, short-term Reversal, and long-term reversal. All returns are in percentages per month.

5 Questions

General remarks

- You can answer Question 1 after Lecture 3, Question 2 after Lecture 5, and Questions 3–5 after Lecture 7.
- In Questions 2–4, you should always use the excess returns of the test assets (so after subtracting the risk-free rate). Factors are already given as excess returns.
- Report returns, pricing errors and volatilities in % with a precision of two decimals.
- The assignment builds heavily on Kelly et al. (2019). Particularly relevant are the Introduction, Sections 2.1, 4.2 and 4.3.

- Seth Pruitt provides a Python package with IPCA on Github and a replication package of the paper on Dropbox.³ You can use one of them or program the routines from scratch.
- Kelly et al. (2019) evaluate the quality of factor models by means of the total R^2 and predictive R^2 . For IPCA in-sample they are defined by eqs. (15) and (16), and for PCA or pre-defined factors in footnote 25.
- Throughout the assignment, only consider IPCA models with $\boldsymbol{\Gamma}_\alpha = \mathbf{O}$.

Question 1

Construct one figure with the efficient frontier with and without the riskless asset based on your 5×5 double-sorted portfolios. Also include the efficient frontier that corresponds with investments in the riskless asset and the market return, the size factor SMB, and the factor that corresponds with your sort (i.e., HML, RMW, CMA, or the factor related to momentum, short-term Reversal, or long-term reversal). To make sure the frontiers are consistent, please proceed as follows

1. For each portfolio, create a series of excess returns by subtracting the risk-free rate (available together with the Fama-French 5 Factors) from the portfolio return at each point in time.
2. Use the excess return series to calculate the excess mean return $\boldsymbol{\mu}^e$ and the variance matrix $\boldsymbol{\Sigma}$.
3. Construct mean returns by adding a fixed value for the risk-free rate. You can fix it at 0.25% per month (corresponding with a gross return of 1.0025).

Use the figure to discuss how efficient the three factors are compared to the 25 portfolios. In particular, compare the expected returns, volatilities and Sharpe ratios of the tangency portfolios.

³Available at <https://github.com/sjpsjpsjp/ipca> and https://www.dropbox.com/sh/sykn8aac1e2s6ry/AAAsGmD_qKRqDp0QhRAyWir_a?dl=0

Question 2

Use GRS-tests to test the CAPM and a three-factor model with the excess market return, the size factor SMB, and the factor that corresponds with your sort. Use the 5×5 double-sorted portfolios as test assets. Report one table with the alphas and their standard errors. Also report the total R^2 and predictive R^2 as defined by Kelly et al. (2019). Discuss the outcome of each GRS-test (you can put the value of the test-statistic and the p -value in a bullet point). Calculate the Sharpe ratio of the market and the Sharpe ratio of the efficient portfolio implied by your three factor model. Use them together with the implications for the Sharpe ratios that follow from the GRS-test to put your results into perspective.

Question 3

Use your complete set of double-sorted portfolio returns to construct factors with PCA and IPCA. First, use both methods to construct a single factor. Then use both methods to construct three factors, and use IPCA to construct five factors. Create a table where you report the alpha for each portfolio i based on each of the PCA and IPCA models (5 models in total), the total and predictive R^2 . Calculate the pricing error for asset i as the average of $r_{it} - z'_{it}\hat{\Gamma}_\beta\hat{\lambda}$, where λ contains the factor risk premia. Compare the results for the different models with each other and with those in the previous question. Does IPCA lead to improvements when you use it construct a single factor? And what about the multi-factor cases? Include in your bullet-list how many characteristics you used, and how you constructed them.

Question 4

Creating factors from and then evaluating them on the same data does not really challenge the method of factor construction much. Therefore, we will now consider a rolling window approach where we use 240 months to construct factors, and then evaluate them on the next return. This works as follows:

1. Use the 240 returns up to and including month t to construct 1 and 3 factors with PCA. Store the resulting orthonormal factor loadings as $\hat{B}_t$ (dimension $N \times K$).

2. Use the 240 returns up to and including month t to construct 1, 3 and 5 factors with IPCA. Store the resulting estimates for $\mathbf{\Gamma}_\beta$ as $\hat{\mathbf{\Gamma}}_{\beta,t}$ (dimension $L \times K$).
3. For both PCA and IPCA, calculate the mean of the resulting factors over these 240 periods and store them in a vector $\hat{\boldsymbol{\lambda}}_t$.
4. Construct the factor realization for $t + 1$. For PCA these are equal to $\mathbf{f}_{t+1} = \hat{\mathbf{B}}'_t \mathbf{r}_{t+1}$, where $\hat{\mathbf{B}}_t$ is from step 1. For IPCA you first calculate predicted loadings $\hat{\mathbf{B}}_{t+1} = \mathbf{Z}_t \hat{\mathbf{\Gamma}}_{\beta,t}$ (this $\mathbf{Z}_t$ has not been used in the estimation), and use them to calculate $\mathbf{f}_{t+1} = (\hat{\mathbf{B}}'_{t+1} \hat{\mathbf{B}}_{t+1})^{-1} \hat{\mathbf{B}}'_{t+1} \mathbf{r}_{t+1}$.
5. Calculate the total-fit residual for the assets as $\boldsymbol{\eta}_{t+1} = \mathbf{r}_{t+1} - \hat{\mathbf{B}}_t \mathbf{f}_{t+1}$ (PCA) or $\boldsymbol{\eta}_{t+1} = \mathbf{r}_{t+1} - \mathbf{Z}_{t+1} \hat{\mathbf{\Gamma}}_{\beta,t} \mathbf{f}_{t+1}$ (IPCA). Also calculate the predicted pricing errors $\boldsymbol{\alpha}_{t+1} = \mathbf{r}_{t+1} - \hat{\mathbf{B}}_t \hat{\boldsymbol{\lambda}}_t$ (PCA) or $\boldsymbol{\alpha}_{t+1} = \mathbf{r}_{t+1} - \mathbf{Z}_{t+1} \hat{\mathbf{\Gamma}}_{\beta,t} \hat{\boldsymbol{\lambda}}_t$ (IPCA) using the values for $\boldsymbol{\lambda}_t$ from step 3.
6. Move one month forward and repeat this procedure until you reach the end of your sample period. This procedure gives you a series of prediction and predicted pricing errors.

Next, you calculate the averages of the series for $\boldsymbol{\eta}_{t+1}$ and $\boldsymbol{\alpha}_{t+1}$ (so for each asset), and you use these series to calculate the total and predictive R^2 .

Also construct error series for the CAPM and the 3-factor model by a reduced procedure. First you regress the portfolio returns on the factors (store the factor loadings as $\hat{\mathbf{B}}_t$), then do step 3 (factor means), and then step 5. You do not need step 4 as the factors are given.

Create a table where you report the average η_{it} and α_{it} for each portfolio based on each of the seven specified models. Compare the results for the different models with each other, and with the in-sample results from the previous question. Does PCA or IPCA lead to improvements when you use it construct a single factor? And what about the multifactor cases?

Question 5

On the last page of the report, formulate your overall answer(s) to the research question “How does IPCA perform relative to standard PCA and *a priori* constructed factor models?”

In your answers, you should connect the different analyses. Think about strengths and weaknesses of your research to give suggestions for further research. Also use this page to reflect on the review you received, and include references, and a declaration of the software and packages you used, including your use of generative AI.

6 The review part of the assignment

Your report should be comprehensible to other students. They should be able to understand your tables and figures, so make sure they have the required information (legends, labels, headings). They should also be able to understand how you reach conclusions based on your results. Therefore, this assignment contains peer review. You will give feedback on the work of other students and receive feedback yourself. You should use it to improve your report.

Giving feedback

You should evaluate the report of another group using the Turnitin environment that is available on Canvas. In general, you should provide feedback on the following points

- Are the results plausible and clearly reported?
- Is it clear how the implications and conclusions follow from the results?

You can also indicate in the report where you spot errors, or have remarks or questions. Feedback should be constructive. Be critical where necessary, but also give praise where deserved!

Receiving feedback

Update your report based on the feedback. On page five, you can reflect on the quality of the received feedback.

7 Requirements for the report and handing in

7.1 Who?

Because Turnitin is not well equipped to deal with student teams, the student with the **lowest** student number should do all submissions (intermediate and final versions, feedback). Only this student can see the report to review, and will also receive the feedback of the other team.

7.2 Intermediate version

The intermediate version should be blind, so do not include your name, start or end dates, or student numbers. On Canvas it is indicated where to hand in the intermediate versions of the assignment. There is also a manual explaining how to give feedback in Turnitin.

7.3 Final version

Your final report (in pdf) should contain the following information

- Names and student numbers.
- Start and end dates of your sample period.
- The number of time-series observations of your sample.
- The filename of the report should consist of the four student numbers, separated by an underscore, i.e. 123456ab_125689rw_321675ek_675412aq.pdf.

You should also submit a zip file with your data and code. Make sure that it is clear what is what. You should use the same file name convention as for the report, i.e. 123456ab_125689rw_321675ek_675412aq.zip.

Canvas contains separate assignments for handing in the final report of the assignment, and the data and code.

8 Evaluation

The evaluation of your report will be based on the same criteria as the feedback. Weights are as follows.

Question	Weight (in %)
1	20
2	20
3	25
4	25
5	10

References

- Cochrane, J. H. (2011). Presidential address: Discount rates. *The Journal of Finance*, 66(4):1047–1108.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
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- Jensen, T. I., Kelly, B., and Pedersen, L. H. (2023). Is there a replication crisis in finance? *The Journal of Finance*, 78(5):2465–2518.
- Kelly, B. T., Pruitt, S., and Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524.